2N5781-2N5786

Silicon N-P-N and P-N-P Epitaxial-Base Complementary-Symmetry Transistors

General-Purpose Types for Switching and Linear-Amplifier Applications

RCA-2N5781, 2N5782, and 2N5783 are epitaxial-base silicon p-n-p transistors - - complements of the hometaxial-base silicon n-p-n types 2N5784, 2N5785, and 2N5786,* respectively.

The three types in each family differ primarily in voltage ratings and saturation characteristics.

These transistors are intended for medium-power switching and complementary-symmetry audio amplifier applications.

* Formerly RCA Dev. Types TA7270, TA7271, TA7272, TA7289, TA7290, and TA7291 respectively.

Features:

- Low saturation voltages
- Maximum safe-area-of-operation curves
- Hermetically sealed package
- High gain at high current
- High breakdown voltages

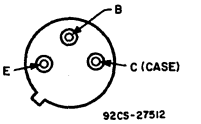
MAXIMUM RATINGS, *Absolute-Maximum Values:*

		P-N-P / N-P-N	2N5781♦ / 2N5784	2N5782♦ / 2N5785	2N5783♦ / 2N5786	
*COLLECTOR-TO-BASE VOLTAGE	V_{CBO}		80	65	45	V
*COLLECTOR-TO-EMITTER SUSTAINING VOLTAGE:						
* With external base-to-emitter resistance (R_{BE}) = 100 Ω	$V_{CER(sus)}$		80	65	45	V
With base open	$V_{CEO(sus)}$		65	50	40	V
*EMITTER-TO-BASE VOLTAGE	V_{EBO}		5	5	3.5	V
*CONTINUOUS COLLECTOR CURRENT	I_C		3.5	3.5	3.5	A
*CONTINUOUS BASE CURRENT	I_B		1	1	1	A
*TRANSISTOR DISSIPATION:	P_T					
At case temperatures up to 25°C			10	10	10	W
At ambient temperatures up to 25°C			1	1	1	W
At case temperatures above 25°C ..... Derate linearly			0.057 W/°C, or see Fig. 1			
At ambient temperatures above 25°C ..... Derate linearly			0.0057			W/°C
*TEMPERATURE RANGE:						
Storage and operating (Junction)			−65 to +200			°C
*LEAD TEMPERATURE (During soldering):						
At distance ≥ 1/32 in. (0.8 mm) from seating plane for 10 s max.			230			°C

*In accordance with JEDEC registration data format JS-6 RDF-2.

♦For p-n-p devices, voltage and current values are negative.

TERMINAL DESIGNATIONS



JEDEC TO-39

(See dimensional outline "C".)

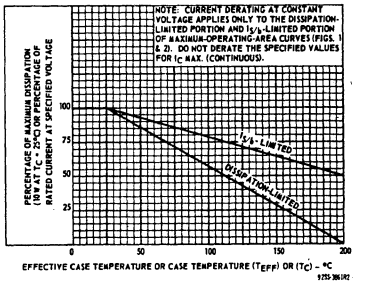


Fig. 1 - Dissipation derating curve for all types.

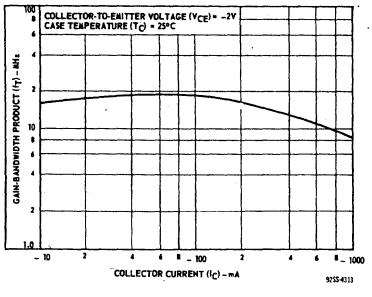


Fig. 2 - Typical gain-bandwidth product for 2N5781, 2N5782, & 2N5783.

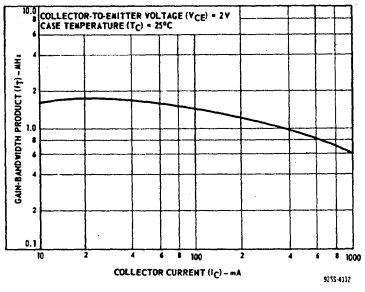


Fig. 3 - Typical gain-bandwidth product for 2N5784, 2N5785, & 2N5786.

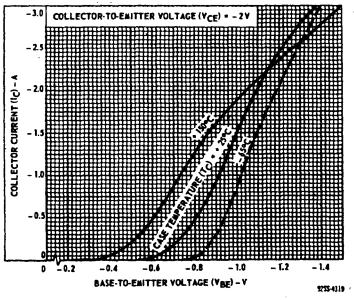


Fig. 4 - Typical transfer characteristics for types 2N5781, 2N5782, 2N5783.

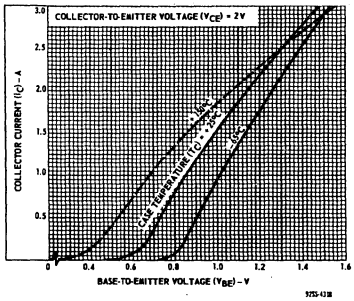


Fig. 5 - Typical transfer characteristics for types 2N5784, 2N5785, 2N5786.

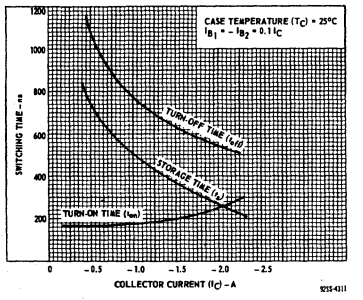


Fig. 6 - Typical saturated switching characteristics for types 2N5781, 2N5782, 2N5783.

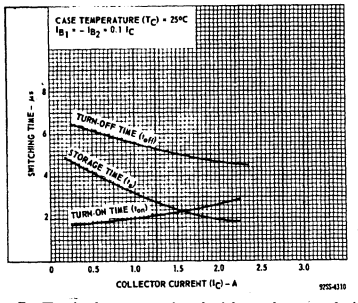


Fig. 7 - Typical saturated switching characteristics for types 2N5784, 2N5785, & 2N5786.

ELECTRICAL CHARACTERISTICS, *At Case Temperature (T_C) = 25°C unless otherwise specified*

CHARACTERISTIC	SYMBOL	TEST CONDITIONS♦ VOLTAGE V dc		CURRENT A dc		LIMITS 2N5781 p-n-p		2N5784 n-p-n		UNITS
		V_{CE}	V_{BE}	I_C	I_B	Min.	Max.	Min.	Max.	
Collector Cutoff Current: With external base-to-emitter resistance (R_{BE}) = 100 Ω	I_{CER}	65				–	–10	–	10	μA
At T_C = 150°C	I_{CER}	65				–	–1	–	1	mA
* With base-emitter junction reverse-biased and external base-to-emitter resistance (R_{BE}) = 100 Ω	I_{CEX}	–75 75	1.5 –1.5			– –	–10 –	– –	– 10	μA
* At T_C = 150°C	I_{CEX}	–75 75	1.5 –1.5			– –	–1 –	– –	– 1	mA
* With base open	I_{CEO}	60			0	–	–100	–	100	μA
* Emitter Cutoff Current	I_{EBO}		–5	0		–	–10	–	10	μA
* DC Forward-Current Transfer Ratio	h_{FE}	2 2		1[a] 3.2[a]		20 4	100 –	20 4	100 –	
* Collector-to-Emitter Sustaining Voltage (see Figs. 2 and 3): With base open	$V_{CEO(sus)}$			0.1[a]	0	–65[b]	–	65[b]	–	V
With external base-to-emitter resistance (R_{BE}) = 100 Ω	$V_{CER(sus)}$			0.1[a]		–80[b]	–	80[b]	–	
* Base-to-Emitter Voltage	V_{BE}	2		1[a]		–	–1.5	–	1.5	V
* Collector-to-Emitter Saturation Voltage (measured 0.25 in (6.35 mm) from case)[c]	$V_{CE(sat)}$			1[a]	0.1	–	–0.5	–	0.5	V
* Magnitude of Common-Emitter, Small-Signal, Short-Circuit, Forward-Current Transfer Ratio[d] f = 4 MHz	$\|h_{fe}\|$	–2		–0.1		2	15	–	–	
f = 200 kHz	$\|h_{fe}\|$	2		0.1				5	20	
* Common-Emitter, Small-Signal, Short-Circuit, Forward-Current Transfer Ratio (f = 1 kHz)	h_{fe}	2		0.1		25	–	25	–	
Saturated Switching Time (V_{CC} = 30 V, I_{B1} = I_{B2}): Turn-on ($t_d + t_r$)	t_{ON}			–1 1	–0.1 0.1	– –	0.5 –	– –	– 5	μs
Turn-off ($t_s + t_f$)	t_{OFF}			–1 1	–0.1 0.1	– –	2.5 –	– –	– 15	
Thermal Resistance: Junction-to-case	$R_{\theta JC}$						17.5	–	17.5	°C/W
Junction-to-ambient	$R_{\theta JA}$					–	175	–	175	

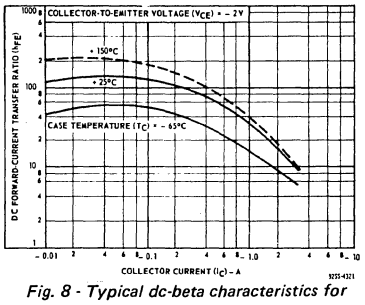


Fig. 8 - Typical dc-beta characteristics for type 2N5781.

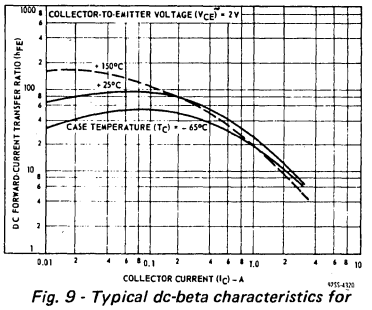


Fig. 9 - Typical dc-beta characteristics for type 2N5784.

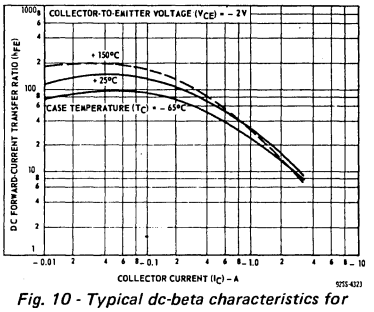


Fig. 10 - Typical dc-beta characteristics for type 2N5782.

2N5781-2N5786

ELECTRICAL CHARACTERISTICS, *At Case Temperature (T_C) = 25°C unless otherwise specified*

	CHARACTERISTIC	SYMBOL	TEST CONDITIONS♦ VOLTAGE V dc V_{CE}	TEST CONDITIONS♦ VOLTAGE V dc V_{BE}	TEST CONDITIONS♦ CURRENT A dc I_C	TEST CONDITIONS♦ CURRENT A dc I_B	LIMITS 2N5782 p-n-p Min.	LIMITS 2N5782 p-n-p Max.	LIMITS 2N5785 n-p-n Min.	LIMITS 2N5785 n-p-n Max.	UNITS
	Collector Cutoff Current: With external base-to-emitter resistance (R_{BE}) = 100 Ω	I_{CER}	50				–	–10	–	10	μA
	At T_C = 150°C		50				–	–1	–	1	mA
*	With base-emitter junction reverse-biased and external base-to-emitter resistance (R_{BE}) = 100 Ω	I_{CEX}	–60 60	1.5 –1.5			– –	–10 –	– –	– 10	μA
*	At T_C = 150°C		–60 60	1.5 –1.5			– –	–1 –	– –	– 1	mA
*	With base open	I_{CEO}	35			0	–	–100	–	100	μA
*	Emitter Cutoff Current	I_{EBO}		–5	0		–	–10	–	10	μA
*	DC Forward-Current Transfer Ratio	h_{FE}	2 2		1.2[a] 3.2[a]		20 4	100 –	20 4	100 –	
*	Collector-to-Emitter Sustaining Voltage (see Figs. 2 and 3): With base open	$V_{CEO(sus)}$			0.1[a]	0	–50[b]	–	50[b]	–	V
	With external base-to-emitter resistance (R_{BE}) = 100 Ω	$V_{CER(sus)}$			0.1[a]		–65[b]	–	65[b]	–	
*	Base-to-Emitter Voltage	V_{BE}	2		1.2[a]		–	–1.5	–	1.5	V
*	Collector-to-Emitter Saturation Voltage (measured 0.25 in (6.35 mm) from case)[c]	$V_{CE(sat)}$			1.2[a] 3.2[a]	0.12 0.8	– –	–0.75 –2	– –	0.75 2	V
*	Magnitude of Common-Emitter, Small-Signal, Short-Circuit, Forward-Current Transfer Ratio[d] f = 4 MHz	$\|h_{fe}\|$	–2		–0.1		2	15	–	–	
	f = 200 kHz		2		0.1		–	–	5	20	
*	Common-Emitter, Small-Signal, Short-Circuit, Forward-Current Transfer Ratio (f = 1 kHz)	h_{fe}	2		0.1		25	–	25	–	
	Saturated Switching Time (V_{CC} = 30 V, I_{B1} = I_{B2}): Turn-on ($t_d + t_r$)	t_{ON}			–1 1	–0.1 0.1	– –	0.5 –	– –	– 5	μs
	Turn-off ($t_s + t_f$)	t_{OFF}			–1 1	–0.1 0.1	– –	2.5 –	– –	– 15	
	Thermal Resistance: Junction-to-case	$R_{\theta JC}$						17.5	–	17.5	°C/W
	Junction-to-ambient	$R_{\theta JA}$					–	175	–	175	

* In accordance with JEDEC registration data format JS-6 RDF-2.

♦ For p-n-p devices, voltage and current values are negative.

[a] Pulsed, pulse duration = 300 μs, duty factor = 1.8%.

[b] *CAUTION: Sustaining voltages $V_{CEO(sus)}$ and $V_{CER(sus)}$ MUST NOT be measured on a curve tracer.*

[c] Lead resistance is critical in this test.

[d] Measured at a frequency where $|h_{fe}|$ is decreasing at approximately 6 dB per octave.

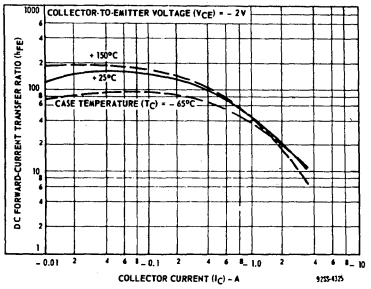


Fig. 11 - Typical dc-beta characteristics for type 2N5783.

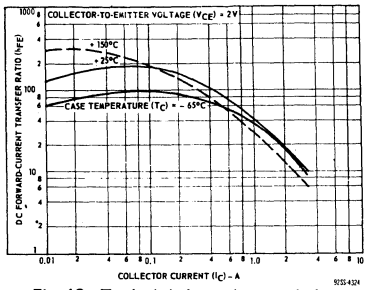


Fig. 12 - Typical dc-beta characteristics for type 2N5786.

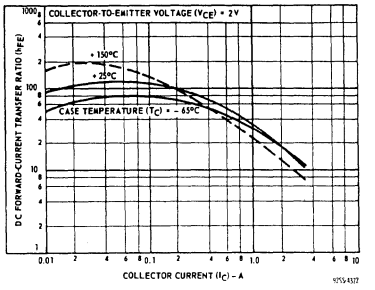


Fig. 13 - Typical dc-beta characteristics for type 2N5785.